

Arboriculture

Science

May, 2026



Short Title: Arboriculture
Faculty Science and Computing
Department Science
Cluster Horticulture
Author EWOODBYRNE
Credits 5

Level: Intermediate

Description of Module / Aims

Arboriculture aims to provide the student with an understanding of the principles and processes involved in the establishment and management of amenity trees.

Programmes

HORT-0043	BSc in Horticulture (WD_SHORT_D)	3 / 6 / M
HORT-0043	BSc in Horticulture (WD_SHORT_DX)	3 / 6 / E

Indicative Content

- Benefits of trees and valuation of trees
- Tree establishment and management; species selection, planting including semi-mature trees, pruning, tree response to wounding
- Decay detection in trees
- Tree pests and pathogens
- Tree hazard assessment and management
- Importance and management of veteran, ancient and heritage trees
- Protection of trees in the development process
- Amenity woodland management

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Debate the benefits of trees and apply a method for valuing amenity trees.
2. Relate the processes involved in tree establishment and maintenance to the development of healthier trees.
3. Compare a range of technologies for detecting decay in trees.
4. Inspect trees for hazards and propose appropriate solutions.
5. Explain the principles of managing trees (including veteran, ancient, and heritage trees), including pest and pathogen impact.
6. Associate the potential impacts of the development process on trees with the strategies used to protect trees in this context.
7. Analyse the importance of amenity woodlands and explain the processes involved in managing these woodlands.

Learning and Teaching Methods

- Lectures
- Fieldwork/tree walks

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3,5,6
Continuous Assessment	30%	
<i>Group Project</i>	10%	7
<i>Project</i>	20%	4

Assessment Criteria

<40%: Little or no knowledge and understanding of the subject matter. Fails to meet project objectives.

40%-49%: Demonstrates a basic level of knowledge and understanding. Project objectives achieved to a basic level.

50%-59%: Demonstrates a satisfactory level of knowledge and understanding. Good attempt to achieve project objectives.

60%-69%: Demonstrates a very good level of understanding, analysis and evaluation. Project objectives achieved.

70%-100%: Demonstrates an excellent level of understanding, analysis and evaluation. Project objectives achieved.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Field Trips	12	
Independent Learning	87	

Essential Material

- Collins, K, ed. *_Amenity Trees and Woodlands: A Guide to their Management in Ireland_*. Dublin: Tree Council of Ireland, 2010.
- Watson, R. *_Trees: Their Use, Management, Cultivation and Biology_*. Wiltshire: Crowwood Press, 2006.

Supplementary Material

- British Standards Institution. *_BS 3998 Tree Work: Recommendations_* London. 2010.
- British Standards Institution. *_BS 5837 Trees in Relation to Design, Demolition and Construction: Recommendations_* London. 2012.